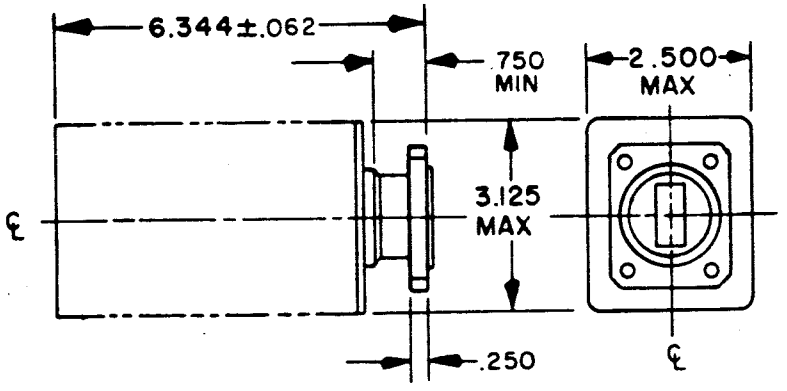
MIL-D-3954/11A
29 May 1974
SUPERSEDING
MIL-D-3954/11
14 October 1966, and
MS35886
15 May 1962

MILITARY SPECIFICATION SHEET

DUMMY LOADS, ELECTRICAL, WAVEGUIDE (FREQUENCY RANGE 7.05 TO 10.0 GIGAHERTZ)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the dummy loads described herein shall consist of this document and the latest issue of Specification MIL-D-3954.



INCHES	MM
.062	1.57
.250	6.35
.750	19.05
2.500	63.50
3.125	79.38
6.344	161.14

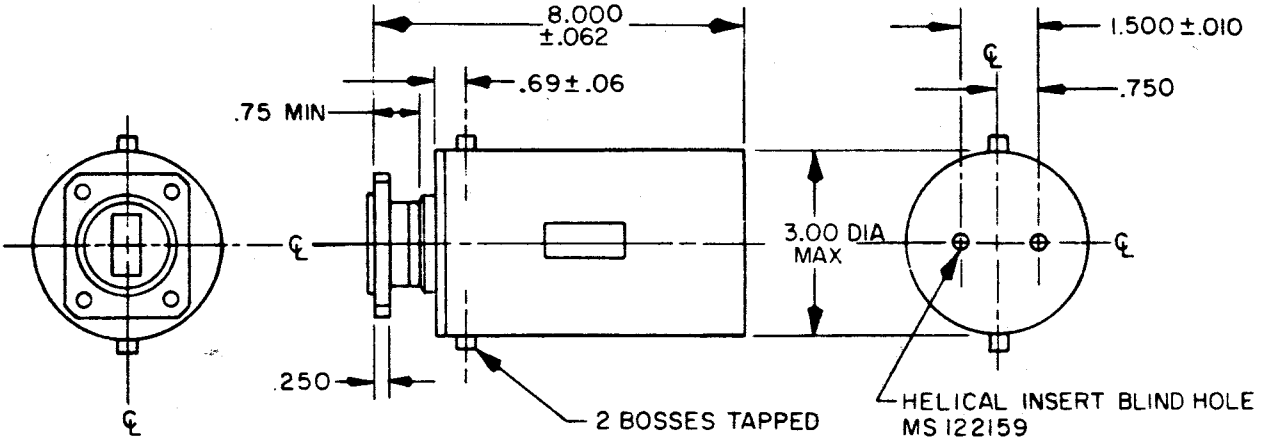
NOTES.
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Screws, lockwashers and gasket are supplied with mating flange.
5. Quantity of fins required will be dependent on heat level requirement of the load.

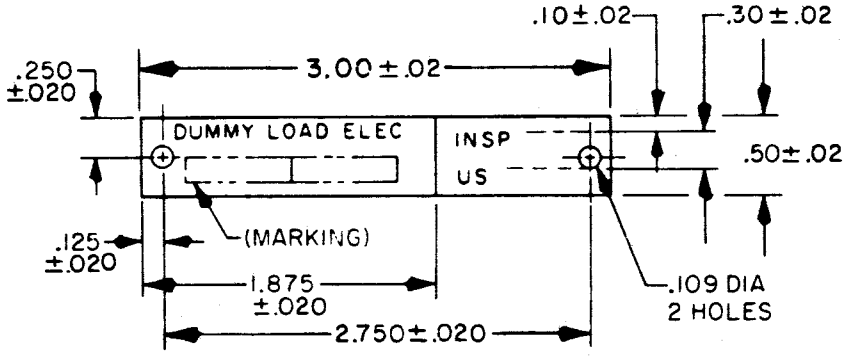
Ⓐ FIGURE 1. Class I dummy load.

Ⓐ denotes changes.

FSC 5985

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IDENTIFICATION PLATE

INCHES	MM
.010	.25
.02	.51
.06	1.52
.062	1.57
.10	2.54
.109	2.77
.125	3.18
.164	4.17
.250	6.35
.30	7.62
.328	8.33
.50	12.70
.69	17.53
.75	19.05
1.500	38.10
1.875	47.63
2.750	69.85
3.00	76.20
8.000	203.20

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Screws, lockwashers and gasket are supplied with mating flange.

Ⓐ FIGURE 2. Class II dummy load.

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TABLE I. Characteristics.

Part number	Class Ⓐ	Material	Flange equal to Ⓐ 2/	VSWR max	Power		Pressure		Figure	Flow rate	Input temp.	Output temp.
					Average	Peak	Internal	Coolant chamber				
					(watts)	(kilowatts)	(psig)	(psig)		(GPM 1/)	(°C)	(°C)
M3954/11-01	I	AL	M3922/53-004 (UG-138/U)	1.10	3/ 500	460	30	---	1			
M3954/11-02	II	AL	M3922/53-004 (UG-138/U)	1.10	1500	460	30	100	2	0.5	35	60
M3954/11-03	II	Copper	M3922/53-002 (UG-51/U)	1.10	1500	460	30	100	2	0.5	35	60
M3954/11-04	II	Corrosion resisting steel	M3922/53-002 (UG-51/U)	1.10	1500	460	30	100	2	0.5	35	60

1/ The flow rate was determined from the following formula:

$$Q = \frac{6.8P}{Cp\ \Delta T}$$

Where: Q = Minimum flow rate in GPM.
P = Avg power in kilowatts.
Cp = Specific heat of coolant.
ΔT = Coolant temperature rise in °F.

The calculations were made for Cp = 1 for water, a T of 45°F and a safety factor of approximately 2. For different coolants of different temperature rises, a different flow rate would be necessary.

2/ Except thickness.

3/ This is a test power value. The average rated power is 600 watts.

Ⓐ TABLE II. Cross reference of part number/ AN nomenclature.

Part number	AN nomenclature
M3954/11-01	DA-148

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REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figures 1 and 2.
Weight:
Dry loads - 1.6 lbs max.
Liquid cooled load - 4.2 lbs max.

Performance characteristics: See table I.

Part number: M3954/11- (dash number from table I).

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - CG, MC, AS, OS, SH
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0754-1)

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